

Elevator Controller

Your task is to create the controller for Elevator in a building of M floors. On every floor there is two call buttons (Up and Down) to call for elevator and the screen showing elevator's current state (floor and direction). In the Elevator there are M buttons, every button is associated with one floor. Your task is to create the controller that takes the passenger to his goal as fast as possible.

Tasks:

- Create the architecture of the solution, which should include all needed classes and threads
- List the hardware functions (the handles – what you give and what you get) you need – for example, GoToFloor, GetCurrentElevatorFloor or others.
- **You can suggest that hardware functions are provided to you, you only need to write their declaration**
- Create algorithm for directing elevator to manage the requests by following logic:
    - Call from any floor, will be treated when Elevator is free (define condition for free Elevator)
    - The call **from inside** of elevator has higher priority than the call **for** the elevator
    - Elevator will stop on floor if someone from inside asked for this floor or if someone outside ask for elevator **in the same direction** the elevator is going
    - Is there some problem with presented logic? Do you have any suggestion how to improve it?
- Write the controller using the logic you wrote and the handles to hardware functions. What threads do you need? What can be done in parallel and what must be implemented asynchronously?
- How would change your solution if there are two elevators in the building? You don't need to write code for this case.

You're welcome to contact us for any question!